Haleh Damirchi

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Machine learning researcher and PhD graduate from Queen's University with expertise in deep learning, probabilistic forecasting, and pedestrian trajectory prediction.

Work Experience

Graduate Research Fellow May 2021 – April 2026
Queen's University *Aiim lab, RCV lab*

- Conducted advanced research in pedestrian trajectory forecasting, counting and localization using deep learning methods.
- Published peer-reviewed research at leading venues including ICASSP and WACV.
- Reported research progress to supervisors on a weekly basis and refined experiments through iterative design and critical analysis of model performance.

University Lecturer Jan 2026 – April 2026
Queen's University *Introduction to Data Science*

- Delivered weekly lectures and provided responsive academic support to a large undergraduate class.
- Led lab sessions and guided students through assignments, projects, and core data science concepts.
- Developed lecture materials and designed assessments aligned with course learning objectives.

Teaching Assistant 2021 – 2025
Queen's University

- Supported instruction in Machine Vision with Deep Learning, C Programming, and Artificial Intelligence courses.
- Evaluated labs, assignments, and exams with careful attention to consistency and academic standards.
- Mentored students through coursework and helped them strengthen debugging and problem-solving skills.

Teaching Assistant 2019 – 2021
Amirkabir University of Technology

- Assisted in teaching Machine Learning and Logic Circuits courses for undergraduate students.
- Designed tutorials, quizzes, and exam materials to reinforce core concepts and improve student understanding.
- Conducted weekly tutorial sessions and provided clear technical guidance in problem solving and coursework.

R&D Intern Summer, Fall 2016
Aria Kavosh Industrial Corp.

- Built and wired an educational control unit integrating relays, PLCs, and an HMI interface.
- Developed detailed wiring and logic diagrams in Microsoft Visio to document system design.

Tech Blogger Aug 2015 – June 2016
Graph Team - ZAMANA blog

- Wrote and edited engaging articles on technology and engineering topics for a broad audience.
- Promoted published content to expand visibility and strengthen audience reach.
- Contributed to broader interest in STEM by creating accessible and inclusive technical content.

R&D Intern Summer 2014
Tabriz Peguh

- Conducted research on a sensor-based system for distinguishing between gasoline and diesel fuel in vehicle fuel tanks.

Qualifications

- Hands-on experience developing, training, and evaluating deep learning models across computer vision, trajectory forecasting, and speech processing.
- Strong programming background in Python, PyTorch, TensorFlow, MATLAB, and C.
- Proven research and communication skills demonstrated through peer-reviewed publications, university teaching, and technical writing.
- Experienced in analyzing model performance, comparing methods rigorously, and translating research problems into practical ML workflows.

Technical Skills

- C, Python, TensorFlow, Keras, PyTorch, Matlab, LaTeX, MySQL
- AVR, PCB Design, PLC Ladder

Publications

[1] **H. Damirchi**, M. Greenspan and A. Etemad, "A Comprehensive and Critical Analysis of Metrics and Evaluation Practices in Pedestrian Trajectory Forecasting", Under Review at NeurIPS 2026.

[2] **H. Damirchi**, A. Etemad and M. Greenspan, "Socially-informed Reconstruction for Pedestrian Trajectory Forecasting", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).

[3] **H. Damirchi**, M. Greenspan and A. Etemad, "Context-Aware Pedestrian Trajectory Prediction with Multimodal Transformer", International Conference on Image Processing (ICIP 2023).

[4] M. Zand, **H. Damirchi** and A. Farley, M. Molahasani, "Multiscale Crowd Counting and Localization by Multitask Point Supervision ", IEEE International Conference on Acoustics (ICASSP 2022).

[5] **H. Damirchi**, S. Seyedin and S. M. Ahadi, "Speaker Extraction Using Stacked BLSTM Optimized with Frequency-domain Differentiated Spectrum Loss", Iranian Conference on Electrical Engineering (ICEE 2020).

[6] **H. Damirchi**, S. Seyedin, S. M. Ahadi, "Improving the Loss Function Efficiency for Speaker Extraction Using Psychoacoustic Effects", *Applied Acoustics.*

Education

Queen's University — Kingston, Canada
PhD. Electrical and Computer Engineering — *2021 – 2026*
Thesis: Pedestrian Trajectory Forecasting in Urban Scenarios

Amirkabir University of Technology — Tehran, Iran
M.Sc. Electrical Engineering — *2017 – 2020*
Thesis: Single-channel Speaker Extraction based on Deep Learning

Tabriz University — Tabriz, Iran
B.Sc. Electrical Engineering — *2012 – 2016*

Projects

Trajectory Forecasting | *Pytorch*

- Proposed novel pedestrian trajectory forecasting solutions using CNNs, RNNs, and Transformers. |
- Conducted a survey on the previous methods in this area to identify research gaps.
- Analyzed evaluation methods for trajectory forecasting both theoretically and empirically to encourage a comprehensive comparison between solutions.

Projects

SlouchLess | *Python*

- An app that tracks camera video to make sure user is not slouching.

Crowd Counting | *Pytorch*

- Proposed a multitask deep neural network using pretrained VGGs for crowd counting and localization.

Facial Recognition | *Pytorch*

- Implemented a facial recognition model using CNNs and facial landmarks as an auxiliary.

Speaker Extraction | *TensorFlow, Pytorch, Matlab*

- Developed models for speaker extraction.
- Proposed novel loss functions for DNN and LSTM models.

Speech Recognition and Enhancement | *Matlab*

- Used Hidden Markov Models (HMM) and Dynamic Time Warping (DTW) to recognize an utterance.
- Enhanced the input noisy speech signal using MMSE and Spectral Subtraction algorithm.
- Applied Ramirez04 algorithm to detect the activity of speech signals.

Backpropagation using Evolutionary Algorithms| *Python*

- Formulated a deep neural network training approach based on Gray Wolf and Particle Swarm Optimization.

Flight Trend Prediction | *Python, TensorFlow*

- Predicted number of flights based on trends and data given by Alibaba company.

Service and Professional Activities

- **Reviewer**: ICLR, NeurIPS 2026; NeurIPS, AISTATS, ICLR, and ICML 2025; NeurIPS and ICASSP 2024; IEEE Transactions on Artificial Intelligence; IEEE Transactions on Affective Computing
- **Guest Lecturer**: Computer Vision with Deep Learning, Queen's University, Fall 2025
- **Organizing Committee Member**: International Conference on Signal Processing and Intelligent Systems, Amirkabir University of Technology, Dec. 2018

Languages

- Azeri, Persian: Native
- English: Fluent